

DEV_IMPL_GUIDE - FOUNDATION_DROOLS

1. Purpose

This guide explains how to implement the Drools/KIE foundation.

Use it with FOUNDATION_DROOLS.md.

This is a provisioning/runtime-setup guide. It is intentionally written as platform setup steps. Business rule execution flows are documented in the relevant workflow implementation guides.

2. What You Are Building

You are deploying a rule execution foundation:

- KIE/Drools runtime.
- Versioned rule packages.
- Rule deployment process.
- Rule execution API contract.
- Monitoring and rollback.

3. Step 1 - Provision Runtime

Medium starting point:

Component	Count
Drools/KIE app nodes	2
DB/repository storage	Foundation DB or approved artifact repository
Load balancer	1 internal endpoint

4. Step 2 - Define KJAR Naming

Example:

```
subscription-workflow-rules
```

Package naming:

```
rules.subscription.common
rules.subscription.activate.wik
rules.subscription.terminate.wik
```

Version every artifact:

```
subscription-workflow-rules:1.0.0
subscription-workflow-rules:1.0.1
```

5. Step 3 - Define Rule API

Recommended request:

```
{
  "packageName": "rules.subscription.activate.wik",
  "requestedDecisions": ["validate-preconditions"],
  "facts": {
    "subscriptionStatus": "CREATED",
    "actorRole": "SALES_AGENT"
  }
}
```

Recommended response:

```
{
  "decisions": {
    "eligible": true,
    "rejectionCode": null
  },
  "rulePackageVersion": "1.0.0"
}
```

}

6. Step 4 - Implement Rule Development Flow

1. Developer edits rules.
2. Unit tests run locally.
3. KJAR builds.
4. KJAR is deployed to test.
5. Integration tests run.
6. KJAR is promoted to staging/production.

Do not hot-edit production rules without versioning and rollback.

7. Step 5 - Rule Authoring Rules

Rules may:

- Read facts.
- Set decisions.
- Reject with reason codes.

Rules must not:

- Call databases.
- Call HTTP APIs.
- Publish Kafka.
- Mutate SOPHIX module state.
- Compute billing money owned by billing.

8. Step 6 - Monitoring

Alert on:

Metric	Alert
Drools node down	Immediate.
Rule execution latency	p95 above SLA.
Rule execution errors	Above threshold.
Unknown package requested	Any sustained count.

9. AWS Option

If using AWS:

- Run KIE/Drools on ECS/EKS.
- Store artifacts in approved artifact repository.
- Store secrets in Secrets Manager/SSM.
- Use private ALB.
- Ship logs to CloudWatch.

10. Tests

Test:

- Known facts return expected decisions.
- Rejection rule returns correct code.

- Package version is visible in response.
- Unknown package fails clearly.
- Rollback to previous package works.

11. Common Mistakes

- Putting DB/API calls in rules.
- Returning free-text errors without machine codes.
- Deploying unversioned rules.
- Using Drools for simple DTO validation.